

Advanced (Habanero)

Q1. Simplify: $\sqrt{16}$

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 10

Q2. Simplify: $\sqrt{25}$

- (A) 3
- (B) 5
- (C) 7
- (D) 9
- (E) 11

Q3. Simplify: $\sqrt{36}$

- (A) 4
- (B) 6
- (C) 8
- (D) 10
- (E) 12

Q4. Simplify: $\sqrt[3]{8}$

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

Q5. Simplify: $\sqrt[3]{27}$

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q6. Convert to radical form: $x^{\frac{1}{2}}$

- (A) $\sqrt{x}$
- (B) $\sqrt[3]{x}$
- (C) x^2
- (D) x^3
- (E) x^4

Q7. Convert to radical form: $x^{\frac{1}{3}}$

- (A) $\sqrt{x}$
- (B) $\sqrt[3]{x}$
- (C) x^2
- (D) x^3
- (E) x^4

Q8. Simplify: $\sqrt{64}$

- (A) 6
- (B) 8
- (C) 10
- (D) 12
- (E) 14

Open Ended Questions (FRQ)

Q9. Simplify the expression: $\sqrt{16x^2}$

Q10. Convert the expression to radical form: $x^{\frac{2}{3}}$

Chef's Tips

- Remind students to check their work
- Encourage students to use a calculator to verify their answers
- Emphasize the importance of showing all work and explaining their reasoning